

Education

- 2007 **B.S. in Computer Engineering - Software**, University of Tehran, Tehran, Iran
- 2010 **M.S. in Artificial Intelligence and Robotics**, University of Tehran, Tehran, Iran
○ Thesis Topic: *Studying the Effect of Structure on Spreading Process in Complex Networks*
○ Adviser: [Professor Masoud Asadpour](#)
- 2017 **Ph.D. in Computer Science**, University of Minnesota, Minneapolis, MN, USA
○ Thesis Topic: *High Dimensional Learning with Structure Inducing Constraints and Regularizers*
○ Adviser: [Professor Arindam Banerjee](#)
- 2021 **Postdoctoral Fellow**, [Mathematical Biosciences Institute](#) and [Department of Biomedical Informatics](#), The Ohio State University, Columbus, OH, USA
○ Mentors: Professors [Kevin R. Coombes](#), [Amanda E. Toland](#), [Sebastian Kurtsek](#), [Tony Nance](#).

Academic Appointments

- 2021–present **Assistant Professor**, [Department of Biostatistics](#) and [Department of Biomedical Informatics](#), Vanderbilt University Medical Center, Nashville, TN, USA

Professional Activities

Intramural

Committees

- 2021–2024 Member of Organizing Committee, *Summer Internship Program for Underrepresented Undergraduates*, Vanderbilt University Medical Center
- 2023–present Member of Admissions Committee, *Biostatistics Graduate Program*, Vanderbilt University

Student Advising

- 2021–2023 Project Advisor, *Yuting Mei*, (Master's student), Data Science, Vanderbilt University
- 2021–2023 Project Advisor, *Chiara Di Gravio*, (Ph.D. student), Biostatistics, Vanderbilt University
- 2022–2023 Thesis Advisor, *Jared Strauch*, (Master's student), Biostatistics, Vanderbilt University
- 2022–2023 Thesis Advisor, *Elisa Yazdani*, (Master's student), Biostatistics, Vanderbilt University
- 2021–2023 Thesis Co-Advisor, *Dora Obodo*, (Ph.D. student), Quantitative & Chemical Biology, Vanderbilt University
- 2023–2024 Project Advisor, *Hao Wang*, (Ph.D. student), Biostatistics, Vanderbilt University

Mentoring

- 2019 Undergraduate Research Mentor (**2** students), REU Program, The Ohio State University
- 2020 Undergraduate Research Mentor (**1** student), Herchel Smith-Harvard Undergraduate Science Research Program, The Ohio State University
- 2022 Undergraduate Research Mentor (**2** students), Summer Internship Program for Underrepresented Undergraduates, Vanderbilt University Medical Center
- 2023 Undergraduate Research Mentor (**2** students), Summer Internship Program for Underrepresented Undergraduates, Vanderbilt University Medical Center

- 2023-2025 Postdoctoral Researcher Mentor, *Leila Azinfar*, R00 HG011367-02 Project, Vanderbilt University Medical Center
- 2024 Undergraduate Research Mentor (1 student), Summer Internship Program for Underrepresented Undergraduates, Vanderbilt University Medical Center
- 2025-present Postdoctoral Researcher Mentor, *Samhita Pal*, ME-2023C1-32148 Project, Vanderbilt University Medical Center

Extramural

Editorial Responsibilities

- 2025-present TOPML: ACM Transactions on Probabilistic Machine Learning *Associate Editor*
- Session Organizer*
- 2022 Vanderbilt Biostatistics Seminar Series
- 2022 CMStatistics2022 session on “Causality and Machine Learning”

Reviewer

- 2019-present BMC Bioinformatics, Journal
- 2020-present BMC Genomics, Journal
- 2021-present PLOS Computational Biology, Journal
- 2022-present PLOS Genetics, Journal
- 2023-present Biometrics, Journal
- 2025-present Nature Communication, Journal

Others

Professional Development

- 2010 Preparing Future Faculty Practicum Course, University of Minnesota
- 2017-2021 Professional Development Seminars, Mathematical Biosciences Institute, The Ohio State University
- 2023-2025 The Geoffrey Fleming Academy for Excellence in Education Mentorship Program (The Fleming Society), Vanderbilt University Medical Center

Panels

- 2018 Panelist at *Career Path Panel* of Summer REU Program: Mathematical Biology Bootcamp, Mathematical Bioscience Institute, The Ohio State University
- 2018 Curriculum development for *Machine Learning for Bioinformatics* course, Department of Biomedical Informatics, The Ohio State University
- 2019 Judge at the *Edward F. Hayes Forum*, The Ohio State University
- 2019 Panelist at *Career Path Panel* of Summer REU Program: Mathematical Biology Bootcamp, Mathematical Bioscience Institute, The Ohio State University
- 2023 Panelist at *Winner Panel* of Bias Detection Tools for Clinical Decision-Making Challenge, NIH’s NCAT
- 2024 Panelist at *AI Discovery and Research Panel* of the launch event of AI Discovery & Vigilance to Accelerate Innovation & Clinical Excellence (ADVANCE) Center, Department of Biomedical Informatics, Vanderbilt University Medical Center

Workshop & Conference Participation

- 2011 Oct. IMA Workshop on Large Graphs: Modeling, Algorithms, and Applications, Minneapolis, MN
- 2012 Mar. IMA Workshop on Machine Learning: Theory and Computation, Minneapolis, MN
- 2012 Oct. 21st Conference on Information and Knowledge Management, Maui, HI
- 2015 Feb. IMA Workshop on Convexity and Optimization: Theory and Applications, Minneapolis, MN

2015 Apr. Analytic Tools in Probability and Applications, Minneapolis, MN
 2015 Apr. IMA Workshop on Information Theory and Concentration Phenomena, Minneapolis, MN
 2015 May. IMA Workshop on Graphical Models, Statistical Inference, and Alg., Minneapolis, MN
 2015 Jul. 31st Conference on Uncertainty in Artificial Intelligence, Amsterdam, Netherlands
 2018 May. Stanford Conference on Big Data in Precision Health, Stanford, CA
 2018 Aug. 34th Conference on Uncertainty in Artificial Intelligence, Monterey, CA
 2018 Jul. CBMS Conference on Elastic Functional and Shape Data Analysis (EFSDA), Columbus, OH
 2018 Oct. Nature Conference on Big Data and Cancer Precision Medicine, Boston, MA
 2018 Oct. MBI Workshop on Math and the Microbiome, Columbus, OH
 2019 Jun. MBI Workshop on Bayesian Causal Inference, Columbus, OH
 2019 Jun. 36th International Conference on Machine Learning, Long Beach, CA
 2019 Jun. Computational Biology Workshop at ICML, Long Beach, CA
 2019 Jun. Adaptive & Multitask Learning: Algorithms & Sys. Workshop at ICML, Long Beach, CA
 2019 Nov. MBI Workshop on Evolutionary Dynamics in Cancer, Columbus, OH
 2020 Jun. 37th International Conference on Machine Learning, Virtual
 2020 Jun. Computational Biology Workshop at ICML, Virtual
 2020 Oct. American Society of Human Genetics, Virtual
 2020 Dec. 34th Conference on Neural Information Processing Systems, Virtual
 2021 Mar. Network Biology, Cold Spring Harbor Laboratory, Virtual
 2021 Apr. 24th International Conference on Artificial Intelligence and Statistics, Virtual
 2021 Apr. Probabilistic Modeling In Genomics, Cold Spring Harbor Laboratory, Virtual
 2021 Apr. American Association for Cancer Research Annual Meeting, Virtual
 2021 Jul. 26th Summer Institute in Statistical Genetics, Department of Biostatistics, University of Washington, Virtual
 2021 Jul. 38th International Conference on Machine Learning, Virtual
 2021 Jul. 37th Conference on Uncertainty in Artificial Intelligence, Virtual
 2021 Dec. 35th Conference on Neural Information Processing Systems, Virtual
 2022 Jul. 27th Summer Institute in Statistical Genetics, Department of Biostatistics, University of Washington, Virtual
 2022 Dec. 36th Conference on Neural Information Processing Systems, Virtual
 2022 Dec. 15th International Conference on Computational and Methodological Statistics, Virtual
 2023 Mar. Network Biology, Cold Spring Harbor Laboratory, Virtual
 2023 Mar. Probabilistic Modeling in Genomics, Cold Spring Harbor Laboratory, Virtual
 2023 Mar. Eastern North American Region (ENAR) Meeting, Nashville, TN
 2023 Apr. Systems Immunology, Cold Spring Harbor Laboratory, Virtual
 2023 Dec. 37th Conference on Neural Information Processing Systems, Virtual

Teaching Experience

The Ohio State University

2018	Machine Learning for Bioinformatics	<i>Co-teaching at the Biomedical Informatics Department</i>
2019	Artificial Intelligence II	<i>Teaching at the Computer Science & Engineering Department</i>
2020	Artificial Intelligence II	<i>Teaching at the Computer Science & Engineering Department</i>

Vanderbilt University

2024-2025	BIOS 7362: Advanced Statistical Inference	<i>Teaching at the Biostatistics Department</i>
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Research Program

- 2020 **K99 HG011367-01 (PI: Asiaee)**, *Causal Effect Estimation of Regulatory Molecules*, NHGRI, \$100k
08/20/2020 - 07/31/2021
- 2021 **R00 HG011367-02 (PI: Asiaee)**, *Causal Effect Estimation of Regulatory Molecules*, NHGRI, \$750k
08/1/2021 - 07/31/2025
- 2024 **ME-2023C1-32148 (PI: Asiaee, Double PI: Huling)**, *Improving Heterogeneous Effect Estimation by Integration of Experimental and Observational Studies*, PCORI, \$750k
02/01/2024 - 01/31/2027

Publications and Presentations

Peer-reviewed Articles

Amir Asiaee T., Mariano Tepper, Arindam Banerjee, and Guillermo Sapiro. If you are happy and you know it... tweet. In *21st ACM international conference on Information and knowledge management (CIKM)*, pages 1602–1606. ACM, 2012.

Amir Asiaee T., Mohammad Afshar, and Masoud Asadpour. Influence maximization for informed agents in collective behavior. In *Distributed Autonomous Robotic Systems*, pages 389–402. Springer, 2013.

Golshan Golnari*, **Amir Asiaee T.***, Arindam Banerjee, and Zhi-Li Zhang. Revisiting non-progressive influence models: Scalable influence maximization in social networks. In *31st Conference on Uncertainty in Artificial Intelligence (UAI)*, pages 316–325, 2015 (*Equal contribution).

Amir Asiaee T., Soumyadeep Chatterjee, and Arindam Banerjee. High dimensional structured estimation with noisy designs. In *16th SIAM International Conference on Data Mining (SDM)*, pages 801–809. SIAM, 2016.

Amir Asiaee T., Hardik Goel, Shalini Ghosh, Vinod Yegneswaran, and Arindam Banerjee. Time series deinterleaving of dns traffic. In *1st Deep Learning and Security Workshop*, 2018.

Zachary B. Abrams, Mark Zucker, Min Wang, **Amir Asiaee Taheri**, Lynne V. Abruzzo, and Kevin R. Coombes. Thirty biologically interpretable clusters of transcription factors distinguish cancer type. *BMC Genomics*, 19(1):738, Oct 2018.

Min Ho* Cho, **Amir Asiaee***, and Sebastian Kurtek. Elastic statistical shape analysis of biological structures with case studies: A tutorial. *Bulletin of Mathematical Biology*, 81(7):2052–2073, 2019.

Amir Asiaee, Samet Oymak, Kevin R. Coombes, and Arindam Banerjee. Data enrichment: Multi-task learning in high dimension with theoretical guarantees. In *Adaptive and Multi-Task Learning Workshop at ICML*, 2019.

Amir Asiaee*, Zachary B. Abrams*, Samantha Nakayiza, Deepa Sampath, and Kevin R. Coombes. Explaining gene expression using twenty-one micrnas. *Journal of Computational Biology*, Forthcoming, 2020 (*Equal contribution).

Sadegh Akhundzadeh, Alireza Omid, Zainab Maleki, Amanda E Toland, Kevin R Coombes, and **Amir Asiaee**. Learning cancer progression network from mutation allele frequencies. In *Computational Biology Workshop at ICML*, 2020.

Phillip B Nicol, Kevin R Coombes, Courtney Deaver, Oksana Chkrebti, Subhadeep Paul, Amanda E Toland, and **Amir Asiaee**. Oncogenetic network estimation with disjunctive bayesian networks. *Computational and Systems Oncology*, 1(2):e1027, 2021.

Phillip B. Nicol, Dániel L. Barabási, Kevin R. Coombes, and **Amir Asiaee**. SITH: An R package for visualizing and analyzing a spatial model of intratumor heterogeneity. *Computational and Systems Oncology*, 2(2):e1033, 2022.

Amir Asiaee, Zachary B Abrams, Heather H Pua, and Kevin R Coombes. Transcriptome complexity disentangled: A regulatory molecules approach. *International Journal of Molecular Sciences*, 26(6):2510, 2025.

Hongxu Zhu, **Amir Asiaee**, Leila Azinfar, Jun Li, Han Liang, Ehsan Irajizad, Kim-Anh Do, and James P Long. AUPRC: a metric for evaluating the performance of in-silico perturbation methods in identifying differentially expressed genes. *Briefings in Bioinformatics*, 26(5):bbaf426, September 2025.

Eric Milan Mukherjee, Dodie Park, **Amir Asiaee**, Matthew S. Krantz, Cosby A. Stone Jr, Michelle Martin-Pozo, and Elizabeth Phillips. Immune checkpoint inhibitors as independent and synergistic drivers of SJS/TEN: An analysis of FAERS. *JAMA Oncology*, 11(12):1542–1545, 2025.

Submitted and Preprints

Jared Strauch and **Amir Asiaee**. Improving drug sensitivity prediction and inference by multi-task learning. *bioRxiv*, 2024.

Dora Obodo and **Amir Asiaee**. Quantifying differential rhythmicity based on effect sizes with limorhyde2. *bioRxiv*, 2024.

Amir Asiaee, Chiara Di Gravio, Cole Beck, Yuting Mei, Samhita Pal, and Jared D. Huling. Improving precision of RCT-based CATE estimation using data borrowing with double calibration. *arXiv*, 2025.

Amir Asiaee, Natalie Mallal, Elizabeth Phillips, and Simon Mallal. Co-evolved partners of immunity: A trait-based map of human keystone organisms. *bioRxiv*, 2025.

Simon Mallal and **Amir Asiaee**. Keystone epitope theory: An ecological perspective on RNA viruses, tumor immunoediting, and vaccine design. *zenodo*, 2025.

Simon Mallal and **Amir Asiaee**. Keystone theory: Implications for effective T and natural killer cell interoperability. *zenodo*, 2025.

Samhita Pal, James O’Quinn, Kaveh Aryan, Heather Pua, James P. Long, and **Amir Asiaee**. DAG DECORation: Continuous optimization for structure learning under hidden confounding. *arXiv*, 2025.

Kaveh Aryan and **Amir Asiaee**. INSITE: Integrating scRNAseq for T cell engineering. *zenodo*, 2025.

Amir Asiaee, Samhita Pal, James O’Quinn, and James P. Long. Causal discovery with mixed latent confounding via precision decomposition. *arXiv*, 2025.

Amir Asiaee and Kaveh Aryan. Fairness under group-conditional prior probability shift: Invariance, drift, and target-aware post-processing. *arXiv*, 2026.

Amir Asiaee and Kaveh Aryan. Fix representation (optimally) before fairness: Finite-sample shrinkage population correction and the true price of fairness under subpopulation shift. *arXiv*, 2026.

Amir Asiaee and Kaveh Aryan. Projected boosting with fairness constraints: Quantifying the cost of fair training distributions. *arXiv*, 2026.

Amir Asiaee, Samhita Pal, Cole Beck, and Jared D. Huling. Sharp bounds for treatment effect generalization under outcome distribution shift. *arXiv*, 2026.

Amir Asiaee, Chao Yan, Zachary B. Abrams, and Bradley A. Malin. PRISM: Differentially private synthetic data with structure-aware budget allocation for prediction. *arXiv*, 2026.

Amir Asiaee, Chao Yan, Zachary B. Abrams, and Bradley A. Malin. Risk-equalized differentially private synthetic data: Protecting outliers by controlling record-level influence. *arXiv*, 2026.

Amir Asiaee. Efficient discovery of approximate causal abstractions via neural mechanism sparsification. *arXiv*, 2026.

Amir Asiaee and Samhita Pal. Noise-calibrated inference from differentially private sufficient statistics in exponential families. *arXiv*, 2026.

Amir Asiaee, Zhuohui J. Liang, and Chao Yan. CausalWrap: Model-agnostic causal constraint wrappers for tabular synthetic data. *arXiv*, 2026.

Amir Asiaee, Kaveh Aryan, and James P. Long. Partial causal structure learning for valid selective conformal inference under interventions. *arXiv*, 2026.

Amir Asiaee and Samhita Pal. Improving RCT-based CATE estimation under covariate mismatch via calibrated alignment. *arXiv*, 2026.

Samhita Pal, Jared D. Huling, and **Amir Asiaee**. Improving RCT-based CATE estimation under covariate mismatch via double calibration. *arXiv*, 2026.

Amir Asiaee, Samhita Pal, and Jared D. Huling. Omitted-variable sensitivity analysis for generalizing randomized trials. *arXiv*, 2026.

Amir Asiaee, Jared Strauch, Leila Azinfar, Samhita Pal, Heather H. Pua, James P. Long, and Kevin R. Coombes. Widespread data leakage inflates performance estimates in cancer drug response prediction. *bioRxiv*, 2026.

Amir Asiaee, James P. Long, Samhita Pal, Heather H. Pua, and Kevin R. Coombes. A shape-constrained regression and wild bootstrap framework for reproducible drug synergy testing. *bioRxiv*, 2026.

Simon Mallal and **Amir Asiaee**. Keystone epitope theory: Implications for the investigation of multiple sclerosis. zenodo, 2026.

Simon Mallal and **Amir Asiaee**. Keystone epitope theory in deep time: How viral endogenization and sexual dimorphism wrote the genomic syllabus of immunity. zenodo, 2026.

Presentations

- 2018 Mar. *Generalized High Dimensional Data Sharing with Application*, MBI Seminar, Columbus, OH, USA
- 2018 Oct. *Inferring Mutations Order from Cross-sectional Cancer Data*, MBI Seminar, Columbus, OH, USA
- 2019 Nov. *Disjunctive Bayesian Network Infers Cancer Progression Network*, Evolutionary Dynamics in Cancer Workshop, MBI, Columbus, OH, USA
- 2019 Nov. *Searching for Effective Combination Cancer Therapy Using Single Doses*, MBI Seminar, Columbus, OH, USA

- 2020 Feb. *Disjunctive Bayesian Network Infers Cancer Progression Network*, Department of Biostatistics, Vanderbilt University, Nashville, TN, USA
- 2021 Oct. *Whole Transcriptome Prediction from Regulatory Molecules*, Bioinformatics Working Group, Department of Biostatistics, Vanderbilt University, Virtual
- 2022 Feb. *Identification and Comparison of Genes Differentially Regulated by Transcription Factors and MicroRNAs*, Department of Biostatistics, University of Louisville, Virtual
- 2022 Dec. *Methods to improve the efficiency of RCTs using observational studies*, Invited Session (Combining Clinical Trials and Observational Study Data) at International Conference on Computational and Methodological Statistics, Virtual
- 2024 Jun. *Leveraging Observational Data for Efficient CATE Estimation in Randomized Controlled Trials*, Invited Session (Advancing Clinical Trial Analysis) at International Chinese Statistical Association, Nashville, TN, USA

Abstracts and Posters

- 2012 Oct. *If You are Happy and You Know It... Tweet*, 21st Conference on Information and Knowledge Management, Maui, HI
- 2015 Jul. *Revisiting Non-progressive Influence Models: Scalable Influence Maximization in Social Networks*, 31st Conference on Uncertainty in Artificial Intelligence, Amsterdam, Netherlands
- 2018 May. *Precision Cancer Treatment with High Dimensional Data Sharing Model*, Stanford Conference on Big Data in Precision Health, Stanford, CA
- 2018 Oct. *Precision Cancer Treatment with High Dimensional Data Enrichment*, Nature Conference on Big Data and Cancer Precision Medicine, Boston, MA
- 2019 Jun. *Explaining Gene Expression Using Twenty-One MicroRNAs*, Computational Biology Workshop at ICML, Long Beach, CA
- 2019 Jun. *Data Enrichment: Multi-task Learning in High Dimension with Theoretical Guarantees*, Adaptive and Multi-Task Learning Workshop at ICML, Long Beach, CA
- 2020 Jun. *Learning Cancer Progression Network from Mutation Allele Frequencies*, Computational Biology Workshop at ICML, Virtual
- 2021 Apr. *Identification and Comparison of Genes Differentially Regulated by Transcription Factors and MicroRNAs*, NHGRI Research Training and Career Development Annual Meeting, Virtual